Blindspot: Helping Assistive Robots to See Partially Hidden People

Software Requirements Specification & Project Plan

1. Document Header & Versioning

Field	Value
Team	Team 11
Supervisor	Dr. Gary Bone, Department of Mechanical Engineering, McMaster University
Version	Version 0.1 Baseline · October 9, 2026
Repository	github.com/richnguyenn/blindspot (official team repository; copy in /docs)

2. Personnel & Roles.

Member	McMaster email	Primary discipline
Kenny Zhao	zhaok31@mcmaster.ca	Project Manager
William Guo	guow42@mcmaster.ca	UI/UX Engineering/Backend Engineering
Eric Park	parke22@mcmaster.ca	Backend Engineering
Bruce Wang	wangb168@mcmaster.ca	Backend Engineering
Richard Nguyen	nguyer22@mcmaster.ca	Applied ML Engineering
Alan Zhou	zhoua39@mcmaster.ca	Applied ML Engineering

3. Definitions, Acronyms & Abbreviations

Term	Meaning
Capsule	Cylinder with hemispherical ends; the output volume for one body segment (15 per person)
Free-space term	Fitting penalty for body parts placed where the sensor saw through to a farther surface
IoU	Intersection over union of two volumes (0–1)
Kinect v2	Microsoft Xbox One Kinect RGB-D sensor, 30 fps, read through libfreenect2
L0–L3	Occlusion levels: share of the camera-facing body hidden (<10%, 10–30%, 30–60%, >60%)
MILO	McMaster Industry Liaison Office (intellectual property)
Occupied-space recall	Share of the true body volume covered by the output capsules
OctoMap	Voxel occupancy-grid library and format used by robot planners
PCD / PCL	Point Cloud Data file format / Point Cloud Library

Term	Meaning
PIPEDA	Personal Information Protection and Electronic Documents Act (Canada)
PoC	Proof of Concept demonstration, Nov 23, 2026
RGB-D	Colour image plus per-pixel depth
ROS 2 / RViz	Robot Operating System 2 / its 3D visualizer
SLAM	Simultaneous localization and mapping (estimates a moving sensor's pose)
SMPL	Skinned Multi-Person Linear model: a parametric 3D human body with pose and shape
WSL2	Windows 10/11 native development environment with vcpkg and Kinect for Windows SDK 2.0

4. Contributions Table

Member	Sections authored or reviewed	Contribution
Kenny Zhao	1, 2, 3, 4, 5, 8	16.7%
William Guo	8, 12, 13, 14	16.7%
Eric Park	9, 15, 16, 17	16.7%
Bruce Wang	6, 7, 10, 11	16.7%
Richard Nguyen	8	16.6%
Alan Zhou	9, 10	16.6%

5. Executive Summary & Purpose

5.1 Problem

Robots operating around humans require reliable location awareness. Standard depth sensors and skeleton trackers often fail during heavy occlusion, non-frontal poses, or brief blockages (e.g., a seated person with legs hidden by a desk). Without anatomically grounded 3D body estimation, robots cannot safely navigate around unobserved limbs.

5.2 Proposed Solution

Blindspot estimates full 3D human volume from a single Kinect v2 stream (30 fps) or recorded PCD files. A C++ core (PCL, Open3D, Eigen) removes background surfaces and isolates subjects. A Python pipeline then processes three layers:

- **Keypoint seeding:** a pretrained 2D pose estimator finds visible joints, which are lifted onto the depth data and labelled as observed or guessed. As a P2 extension, a small trained network predicts the positions of hidden joints from the visible ones to give the body fit a better starting point.
- **Body fitting with free space:** the SMPL body model is fitted to the visible points within joint limits and a shape prior, and is kept out of space the sensor saw through, which constrains where hidden limbs can be.
- **Tracking:** person identities and volumes persist through brief full occlusions, and fits are warmly started from the previous frame.

The system outputs 15 conservative, labelled capsules per person in metres with configurable safety margins. When confidence is low, volumes expand rather than shrink to preserve safety.

5.3 Intended Impact

The immediate impact is safer navigation and manipulation around seated and partially hidden people, such as older adults at a table.

6. Stakeholders & User Personas

Stakeholder	Type	Expectation
Dr. Gary Bone (Mechanical Engineering)	Primary client, supervisor	Works live from a Kinect and offline on PCD files; tested from nearly visible to several people occluded by objects and each other; C++ and/or Python, ideally PCL and Eigen
Capstone team (6 students)	Developers	Feasible scope, clear requirements, working PoC on Nov 23
Course instructor and TA	Evaluators	Complete, testable SRS with quantitative metrics
Researchers in Dr. Bone's lab	Future maintainers	Documented code and a stable interface for a mobile manipulator stack
Assistive-robot users and bystanders	Indirect, safety-affected	The robot never plans through a body part it cannot see

Persona A, robotics researcher (direct user): runs live or recorded input; needs per-frame volumes, reproducible offline runs, a viewer and low-confidence warnings.

Persona B, motion-planning engineer (integrator): needs a versioned output format, predictable latency and conservative volumes.

Persona C, seated home occupant (indirect): needs the robot to track their hidden legs and torso while a caregiver walks past.

7. Constraints & Assumptions

- **Sensor:** one Kinect v2 (30 fps colour point clouds, front view only, usable range about 1–4 m). Depth noise grows with distance; glass and shiny surfaces leave holes.
- **Compute and cost:** the reference platforms are Windows 10/11 laptops equipped with local GPUs. The pipeline runs locally on CPU/GPU; no cloud budget is assumed, and all software is free.
- **Platform:** Windows 10/11 native is the target platform. Live capture and processing run natively on Windows laptops using Kinect for Windows SDK

2.0. Offline PCD work is fully supported on Windows. No robot hardware is guaranteed, so sensor motion is replayed from recordings.

- **Toolchain:** C++17 (MSVC) and Python 3.11, PCL 1.14 and Eigen 3.4 managed via vcpkg, built with CMake; offline PCD playback needs no Kinect driver.
- **Architecture boundary:** the system outputs volumes; robot motion planning and control, identifying people, and training large neural networks are out of scope. The only trained component is the small hidden-joint predictor (FR-SEED-3, P2).
- **Intellectual property:** the project has no IP transfer and is not confidential. Members keep ownership under McMaster's student IP policy; the MILO position and the repository licence (MIT) are confirmed with Dr. Bone in writing before any public release.
- **Licences:** PCL (BSD) and Eigen (MPL2) are open source. SMPL, the pose-estimator weights and any public dataset adopted (AMASS, PROX) are used under their academic licences and are never redistributed (NFR-SP-5).
- **Privacy:** Point clouds are PIPEDA-regulated. Testing uses consenting members; data is stored securely on OneDrive and deleted at project end.

Assumptions: Testing occurs in indoor scenes with up to three people within 0.5–4.5 m using a single static or slowly moving sensor. Targets are detailed in Section 10.2.

8. Functional Requirements

Priorities: **P0** Must Have (core MVP, shown at the Nov 23 PoC) · **P1** High Priority (final system, code freeze Mar 26) · **P2** Medium Priority (if time permits) · **P3** Low Priority (stretch). IDs are FR-<area>-<n>. Each row gives input → output, data-flow stages (Figure 1), whether it is frontend (FE: viewer, command line) or backend (BE: pipeline), and the acceptance threshold. Metrics are defined in 9.3. P0 thresholds are checked at the PoC and must still hold at code freeze; P1 thresholds apply at code freeze.

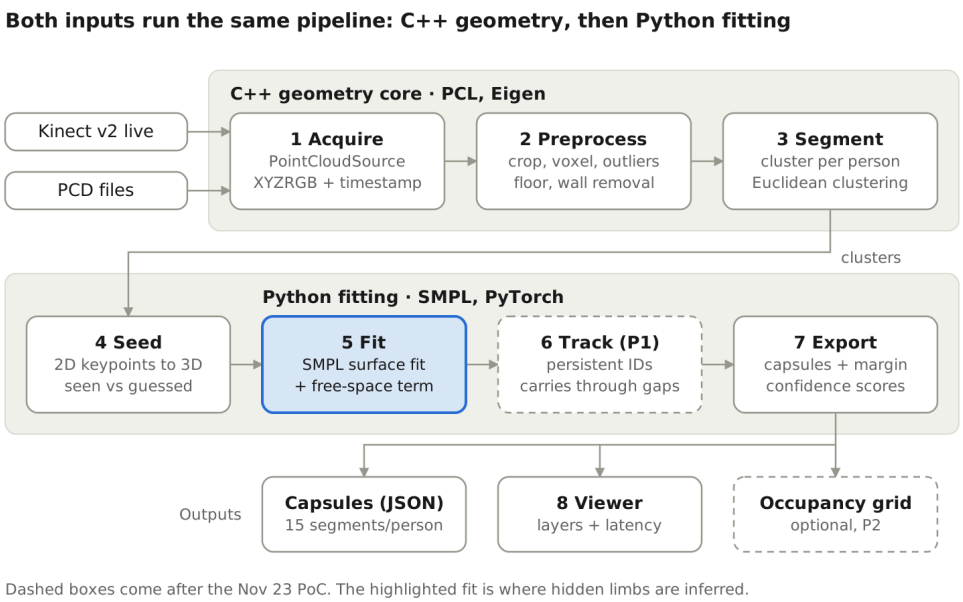


Figure 1. Pipeline stages 1–8 (dashed stages come after the PoC).

8.1 P0 – Must Have

Requirement	Input → output	Flow · FE/BE	Acceptance
FR-ACQ-1 Read PCD files and live Kinect v2 frames via Kinect for Windows SDK 2.0 through one interface	PCD or Kinect → XYZRGB cloud + timestamp	1→2 · BE	Live ≥25 fps mean over 60 s, ≤2% dropped; all D4 files load with no driver
FR-ACQ-2 Record live streams as replayable PCD sequences	Live stream → timestamped PCDs	1→disk · BE	Same frame count; timestamps within 1 ms
FR-PRE-1 Crop, downsample, remove outliers, floor and walls	Raw cloud → foreground cloud, depth map	2→3, 5 · BE	Person-point retention ≥0.95; ≥95% of floor/wall removed
FR-SEG-1 Isolate one person as a cluster	Foreground → person cluster	3→4 · BE	Point precision and recall ≥0.90 on D1
FR-SEED-1 Detect 2D keypoints, lift to 3D, label observed or guessed	Colour image, cluster → 3D keypoints + confidence	4→5 · BE	Median keypoint error ≤6 cm; hidden-joint labelling ≥0.80
FR-FIT-1 Fit SMPL as a closed body within joint limits and a shape prior	Cluster, keypoints → SMPL pose and shape	5→7 · BE	At L0: IoU ≥0.50, per-limb error ≤8 cm, no joint-limit violations

Requirement	Input → output	Flow · FE/BE	Acceptance
FR-FIT-2 Keep the body out of observed free space	Depth map, SMPL → occlusion-consistent fit	5→7 · BE	Table case (L1): IoU and recall both higher than without the term
FR-OUT-1 Output 15 capsules per person with a configurable margin; never shrink hidden segments	SMPL body, margin → capsules (JSON)	7→disk, 8 · BE	Valid schema; margin +m raises every radius by m (±1 mm); over-coverage ≤2.0
FR-VIZ-1 Viewer with layer toggles, margin slider, per-stage latency	Stage outputs → interactive 3D view	1–7→8 · FE	Every PoC checklist item in Section 16 passes
FR-EVAL-1 Score IoU, recall and per-limb error in CI	Outputs, ground truth → report, CI result	7→report · BE	Recall ≥0.90 at L0; a fit shifted 10 cm fails CI
FR-EVAL-2 Build ground truth from paired unblocked/blocked captures	Paired captures → reference volumes	capture→report · BE	Pose-hold drift ≤2.0 cm RMS

8.2 P1 – High Priority

Requirement	Input → output	Flow · FE/BE	Acceptance
FR-SEG-2 Separate up to 3 people, including touching people	Foreground, keypoints → cluster per person	3→4 · BE	Point assignment ≥0.90 separated, ≥0.80 touching
FR-FIT-3 Fit under heavy object and mutual occlusion	Partial clusters, depth map → full-body fits	5→7 · BE	IoU ≥0.50 / 0.40 / 0.30 at L1 / L2 / L3; recall ≥0.95 at L1–L2, ≥0.90 at L3
FR-FIT-4 Standing, seated, walking, reaching; facing, side-on, turned away	Clusters in varied poses → fit per pose	5→7 · BE	IoU ≥0.50 and recall ≥0.95 in every pose × orientation cell at L0–L1
FR-TRK-1 Keep IDs across frames; predict volumes through brief full occlusion	Current and previous fits → IDs, predicted volumes	5→6→7 · BE	Zero ID switches in ≥90% of crossing sequences; no track lost for occlusions ≤2 s; recall ≥0.90 while hidden

Requirement	Input → output	Flow · FE/BE	Acceptance
FR-OUT-2 Per-segment confidence; enlarged envelope and warning when low	Fit residuals → confidence, envelope	5→7 · BE	Envelope recall ≥0.98; ≥90% of segments with error >10 cm flagged
FR-IO-1 Deterministic command-line batch mode; recover from bad files and sensor loss	Arguments, config → output folder, error log	user→1–7→ disk · FE	Byte-identical JSON over 3 seeded reruns; every injected fault logged without a crash
FR-EVAL-3 Synthetic SMPL scenes with exact ground truth; report by occlusion, people, pose	SMPL bodies, scenes → PCDs, report	generator→ 1, report · BE	≥20 synthetic frames in every P0–P1 cell of 9.2; real data at every P0–P1 occlusion level

8.3 P2 – Medium and P3 – Low Priority

Requirement	Pri.	Flow · FE/BE	Acceptance
FR-OUT-3 Occupancy grid (occupied/free/unknown) in a robot or world frame	P2	7→disk · BE	Capsule-interior cells occupied; ≤1 cell (5 cm OctoMap resolution) error under a known transform
FR-SEED-2 Fall back to geometry cues or the previous track when keypoints fail	P2	3, 6→4 · BE	Fit produced in every such frame; recall ≥0.90
FR-TRK-2 Merge views from a moving sensor (poses from RGB-D SLAM)	P2	1→5 · BE	Multi-view IoU at least 0.05 above single view
FR-VIZ-2 Export screenshots and video in a geometry-only privacy mode	P2	8→disk · FE	Exports byte-identical when input colours are randomised
FR-SEED-3 Predict hidden joint positions from the visible keypoints with a small trained network, and use them to initialise the SMPL fit	P2	4→5 · BE	Median hidden-joint error ≤10 cm on held-out D3 at L1–L2; FR-FIT-3 IoU at L2 at least 0.05 higher with the predictor than without
FR-INT-1 Publish capsules and grid from a ROS 2 node	P3	7→ROS 2 · BE	All 15 capsules shown in RViz; topic rate within 10% of pipeline rate
FR-FIT-5 Lying, crouching and bending poses	P3	5→7 · BE	IoU reported; target ≥0.40

Requirement	Pri.	Flow · FE/BE	Acceptance
FR-TRK-3 Predict each person's volume 0.5 s ahead	P3	6→7 · BE	Recall ≥0.90 of the future body at the default margin

9. Data Strategy & Quantitative Metrics

9.1 Datasets

ID	Dataset and source	Ground truth	Licence	Size (PoC / final)
D1	Labelled real Kinect v2 frames	Hand-labelled person points	Team-owned, consent terms (7)	50 / 200 frames
D2	Paired real captures: the single Kinect v2 live feed records a held pose unblocked, then with the occluder placed (FR-EVAL-2)	SMPL fit to the unblocked capture	Team-owned, consent terms (7)	5–10 sequences (≥2 table cases) / 40+
D3	Synthetic: SMPL bodies posed from motion capture (candidate: AMASS), rendered as virtual Kinect v2 clouds with noise and occluders	Exact mesh and per-joint visibility	AMASS and SMPL academic licences; not redistributed	none / ≥20 frames per cell
D4	CI fixtures drawn from D3 only, in /data/samples (Git LFS)	Inherited from D3	As D3; generated, no real people	3 / 10 fixtures of ≤5 frames, ≤200 MB
D5	Optional real recordings among furniture (candidate: PROX, Kinect v2)	Fitted SMPL-X; visible regions only	Non-commercial research; not redistributed	if adopted: 1–2 scenes / full set

- **Capture:** 0.5–4.5 m, 30 fps clips of ≤30 s (about 3.5 MB per frame, about 6 GB per minute). Each sequence stores scene ID, test-matrix cell, sensor pose, date and consent record ID.
- **Labelling:** masks drawn on the colour image, projected onto the cloud; 10% double-checked.
- **Splits:** 70% development, 30% held-out test, split by session; the test set is frozen at the Jan 25 milestone and never used for tuning.
- **No training set:** the keypoint model is pretrained and is not retrained. FR-SEED-3 trains only on a separate D3 training split, built from AMASS subjects that never appear in the test set. Hidden joints are masked using the same occlusion levels as 9.2, so labels are exact and no hand labelling is needed. The held-out test set frozen on Jan 25 is never used for training or tuning.

- **Candidate sources:** AMASS (for D3) and PROX (D5) are options under evaluation.

9.2 Test Matrix

Dimension	Levels (priority)
Occlusion	L0, L1 (P0) · L2, L3 (P1)
People in scene	1 (P0) · 2, 3 (P1)
Pose	standing (P0) · seated, walking, reaching (P1) · lying, crouching, bending (P3)
Orientation	facing 0° (P0) · side-on 90°, turned away 180° (P1)
Distance	1–2 m (P0) · 2–3 m, 3–4 m (P1)
Sensor motion	static (P0) · moving, poses from RGB-D SLAM (P2)

9.3 Metric Definitions

Volumes are compared on a 2 cm voxel grid in the sensor frame (Open3D). P is the fitted SMPL body, G the ground-truth body, C the capsules at the default margin m_o = 5 cm and C_o the capsules at margin 0.

Metric	Definition
Volumetric IoU	vol(P ∩ G) / vol(P ∪ G)
Occupied-space recall (primary)	vol(C ∩ G) / vol(G)
Over-coverage ratio (guard)	vol(C_o) / vol(G); stops recall being met with oversized capsules
Per-limb error	Mean 3D distance between predicted and true endpoints of the 15 segments (cm)
3D keypoint error	Distance from a lifted observed keypoint to the true joint (cm)
Hidden-joint labelling	Share of truly hidden joints labelled guessed
Person-point retention	Share of labelled person voxels kept after preprocessing
Point precision / recall; assignment	Cluster points vs. hand labels; share assigned to the right person
Pose-hold drift	RMS distance of visible points between the blocked and unblocked capture (cm)
ID switches	Times a person's track ID changes per sequence
Hidden-joint error	Distance between a predicted hidden joint and its true position (cm)

9.4 Applying the Thresholds

- Results are averaged over frames, then sequences, and reported as a mean with a 95% confidence interval; a threshold is met when the mean meets it in every required cell.

- CI (FR-EVAL-1) enforces P0 thresholds from the start; P1 rows join CI as each feature merges.
- At the PoC, evaluation uses D2 development data. Full held-out D2 and D3 evaluation runs at the Jan 25 Design + V&V milestone and at code freeze (Mar 26). D2's own fitting error is measured on D3 and reported alongside.
- Thresholds are provisional until the first captures (Oct 23) and the mid-November supervisor check-in; any change is versioned in this SRS (R10).

10. Non-Functional Requirements

ID	Requirement	Pri.	Target / verification
NFR-US-1	A teammate or TA who did not write the code can run the offline pipeline on Windows from the README	P0	≤30 min setup using vcpkg and CMake on Windows, excluding downloads
NFR-US-2	Viewer is responsive and accessible (FR-VIZ-1)	P0	Margin slider updates within 200 ms; every toggle has a key; confidence colours are colour-blind safe (WCAG 2.1 contrast)
NFR-US-3	Integrators can use the output without reading code	P0	Versioned JSON schema documents units, frame and every field
NFR-US-4	Command line usable without the source; low confidence visible (FR-IO-1, FR-OUT-2)	P1	--help covers every option; errors name cause and fix; flagged segments warned in viewer and JSON
NFR-PF-1	Per-stage latency measured every frame	P0	Median and 95th percentile logged and shown
NFR-PF-2	CI regression stays fast	P0	≤15 min on a Windows GitHub-hosted runner
NFR-PF-3	C++ geometry core (stages 1–3) keeps up	P1	≥15 fps with one person
NFR-PF-4	Full pipeline near real time	P1	≥5 fps with one person, ≥3 fps with three; 95th-percentile frame-to-capsule latency ≤300 ms
NFR-PF-5	Memory fits Windows laptops	P1	≤4 GB peak with three people
NFR-SP-1	Raw recordings stay on access-controlled storage	P0	McMaster OneDrive only, shared with team and supervisor; deleted at project end

ID	Requirement	Pri.	Target / verification
NFR-SP-2	No recording of a real person is committed to GitHub	P0	D4 is synthetic; a CI check rejects files without a synthetic-source tag
NFR-SP-3	Every real recording is traceable to consent	P0	Manifest check rejects files without a consent record ID
NFR-SP-4	Runs fully offline	P0	No network calls or telemetry at runtime
NFR-SP-5	Licensed assets are not redistributed	P0	SMPL files and weights fetched by a setup script after licence acceptance; absent from repo and image
NFR-RL-1	Failures err toward occupied space	P0	A failed frame is marked invalid and its space reported as unknown, never free
NFR-RL-2	No person silently dropped	P0	Every frame with a person cluster yields 15 capsules or an error record
NFR-RL-3	Fitting is time-bounded	P0	Iteration cap with flag; ≥95% of evaluation frames converge
NFR-RL-4	Long runs are stable; faults recovered (FR-IO-1)	P1	30 min with no crash and ≤10% memory growth; resumes ≤5 s after sensor reconnects
NFR-MP-1	Builds on Windows 10/11	P0	CMake with MSVC on Windows 10/11 using vcpkg manifest mode for dependencies
NFR-MP-2	Each stage sits behind a documented interface	P0	Swappable stages (e.g. PointCloudSource); Doxygen or docstrings on public APIs

11. Risk Management & Feasibility Analysis

#	Risk	L / I	Mitigation
R1	Hidden-limb volumes inaccurate under heavy occlusion	H / H	Stage occlusion levels; free-space term; conservative margins; report per level
R2	Too little varied test data	H / H	Record from October; ask Dr. Bone's lab for PCD files; D3 synthetic scenes; optional public recordings (D5, 9.1)
R3	Kinect driver fails on a member's machine	M / H	Offline PCD is the main path; native Windows Kinect v2 SDK on laptops; vcpkg dependency management

#	Risk	L / I	Mitigation
R4	Pretrained keypoint detector fails under occlusion or lighting	M / H	Confidence labels; FR-SEED-2 fallback; test across lighting
R5	Touching people merged or swapped	M / H	Dedicated pair fixtures; tracking (FR-TRK-1); lower 0.80 target for touching
R6	Real-time target missed	M / M	Voxel downsampling, cropping, warm starts, early profiling; port fit to C++ only if needed
R7	C++/Python interface churn	M / M	Exchange PCD files first, then pybind11 once interfaces settle
R8	Privacy or consent problem	L / H	Record consenting team members only in initial testing; private storage; confirm ethics with Dr. Bone
R10	Coordination gaps or a member unavailable	L / M	Weekly sync with minutes; shared module ownership; reviews per Section 13
R11	The predictor trained on synthetic data does not transfer to real Kinect data	M / L	Train with simulated Kinect noise; check on D2; P2, so it is dropped if it does not help

12. Team Communication & Meeting Cadence

- **Channels:** Discord day to day; Microsoft Teams for course announcements and sign-ups; McMaster email with the supervisor and TA. Escalation: Team Coordinator → TA → instructor.
- **Weekly sync:** 60 min, Tuesdays at 6:00 PM, in person or on Discord voice; every second week it is the sprint review and planning. Async stand-ups on Discord every Monday and Thursday; class time (Mon/Wed/Thu 1:30–2:20 PM) is co-working time.
- **Records:** every meeting goes in /meetings on GitHub as YYYY-MM-DD-<type>.md (attendees, decisions, owned action items), committed by a rotating scribe within 24 h.
- **Mandatory check-ins** (booked a week ahead): supervisor twice per semester (mid-Oct SRS feedback, mid-Nov PoC dry run; mid-Jan design review, mid-Mar readiness); TA once per semester (late Oct, Feb).

13. Engineering Workflow & Version Control

- **Repository:** /src (io, preprocessing, segmentation, body_model, occlusion, viz), /tests, /data/samples (synthetic fixtures, Git LFS), /docs (all deliverables), /meetings, README.md.
- **Branching:** simplified GitFlow. main is protected, demo-ready and tagged at each milestone (v0.5-poc, v0.8-design, v1.0-final); develop is the integration

branch; work happens on feature/<issue#>-<desc> branches; nobody pushes directly to main or develop.
- **Pull requests:** each links an issue and the FR it addresses; one approval from a non-author to develop, two into main; CI must pass; squash-merge.
- **CI/CD (GitHub Actions, every PR):** CMake build, clang-format and ruff, GoogleTest and pytest with coverage, and the FR-EVAL-1 regression on /data/samples. Each tagged release publishes a Docker image and Linux binaries.
- **Agile:** two-week Scrum sprints Oct 12–Apr 5 (paused for December exams) at about 10 h per member per week. A GitHub Projects board (Backlog → To Do → In Progress → Review → Done) is shared with the TA and instructor; issues carry an owner, story points and a task-P0...task-P4 label (course scale), distinct from requirement priorities. The Coordinator orders the backlog by priority and dependency. Done = merged, tested, documented. Milestones: PoC (Nov 23), Design + V&V (Jan 25), Final (Apr 5).

14. Infrastructure & Compute Resources

- **Development and demo:** one Windows laptop acts as a dedicated shared/remote machine (functioning like a VM) with the Kinect v2 connected via Kinect for Windows SDK 2.0; team members can remote into it for shared development, live capture, and calibration. (Stack used: vcpkg to pin PCL 1.14, Eigen 3.4, Open3D, CMake, MSVC and Python).
- **Storage and CI:** raw PCD data on McMaster OneDrive; synthetic fixtures in Git LFS; Windows GitHub-hosted CI runners.
- **Contingency compute:** if full-dataset batch evaluation of D3 outgrows local machines, the team requests CAS virtual servers or GPU cluster access through the instructor or TA.
- **Distribution:** a desktop/command-line Windows application that runs offline on PCD files or live with a Kinect, shipped as Windows executables via GitHub Releases; no app store or web hosting is needed.
- **Budget:** all software is free. The CAS $125 student hardware reimbursement (TA approval) covers cables, a USB 3 adapter and a tripod; CAS covers Expo poster printing up to $60.

15. Technology Stack & Architectural Rationale

C++ handles fast geometry, where PCL and Eigen are native; Python handles fitting, experiments and evaluation, where iteration speed matters more. The two exchange PCD files first, then pybind11 bindings once interfaces settle.

Layer	Choice	Why
Core / research languages	C++17 / Python 3.11	Native PCL and Eigen; Python is home to SMPL, PyTorch and evaluation tools

Layer	Choice	Why
Sensor	Kinect v2 via Kinect for Windows SDK 2.0 behind PointCloudSource	Live and recorded input share one pipeline, so CI needs no hardware
Point clouds / geometry	PCL / Eigen	Voxel filters, outlier removal, RANSAC planes, Euclidean clustering; transforms and PCA
Body model and fitting	SMPL (smplx) in PyTorch, Adam/L-BFGS	Differentiable model, so surface, free-space and joint-limit terms optimise directly; no model port while the method changes
Keypoint seeding	Pretrained 2D estimator (e.g. RTMPose) on ONNX Runtime	No training; per-joint confidence separates seen from guessed joints
Hidden-joint predictor	Small multilayer perceptron in PyTorch, trained on D3 joint sets	Takes joint coordinates only, so it trains on a laptop CPU; synthetic data gives exact labels
Viewer / output	Open3D / capsules + optional OctoMap	Fast interactive 3D; cheap collision checks; grid suits planners
Testing	GoogleTest/GoogleMock, pytest + pytest-cov, versioned fixtures	Unit tests per stage with a mocked sensor; whole-pipeline threshold checks
Build and CI	CMake, vcpkg, MSVC, GitHub Actions (windows-latest)	Standardized C++ dependency management on Windows via vcpkg; every PR runs tests and coverage on Windows

Rejected alternatives: Hand-built skeletons (lacks proportions) and Ceres (requires SMPL port).

16. Proof of Concept Demonstration Plan (Nov 23)

The demo runs the full pipeline end to end on one person, including one occluded case scored against real ground truth. It tests four risky assumptions: the Kinect and preprocessing chain is reliable; 2D keypoints lift accurately onto depth; SMPL fitting converges on partial clouds; and the free-space term measurably improves hidden-limb estimates.

- **Shown:** (1) one pipeline on the live Kinect feed and a recorded PCD file; (2) floor/wall removal and person isolation; (3) keypoints coloured by confidence; (4) the SMPL body and capsules on the points; (5) a person behind a table, fitted with and without the free-space term and scored against the unblocked capture of the same pose (D2); (6) a CI run reporting IoU and recall.

- **Interactive features:** switch live/file input; toggle raw, segmented, keypoint, body and capsule layers; margin slider; per-stage latency overlay.
- **Out of scope for the PoC:** multiple people, unusual poses, C++ port of the fit, ROS 2.
- **Plan:** Sep 28–Oct 11 Windows vcpkg environment, CI skeleton, Kinect SDK bring-up · Oct 5–23 C++ preprocessing and segmentation, first 5–10 paired captures · Oct 19–Nov 6 keypoint seeding, SMPL fit on visible people · Oct 26–Nov 18 free-space term, table case, evaluation harness · Nov 16–22 integration, two dry runs, backup video.
- **Success criteria:** runs on live and recorded input; on the table case IoU and recall both improve with the free-space term (FR-FIT-2); latency reported as measured; CI green.

17. Project Schedule & Gantt Chart

Fall proves the single-person pipeline by the Nov 23 PoC; Winter adds heavy occlusion, multiple people (from late January) and more poses. Milestones: Design + V&V Jan 25, code freeze Mar 26, final deliverables Apr 5, Expo early April (Apr 8 placeholder). Owners are shown beside each bar.

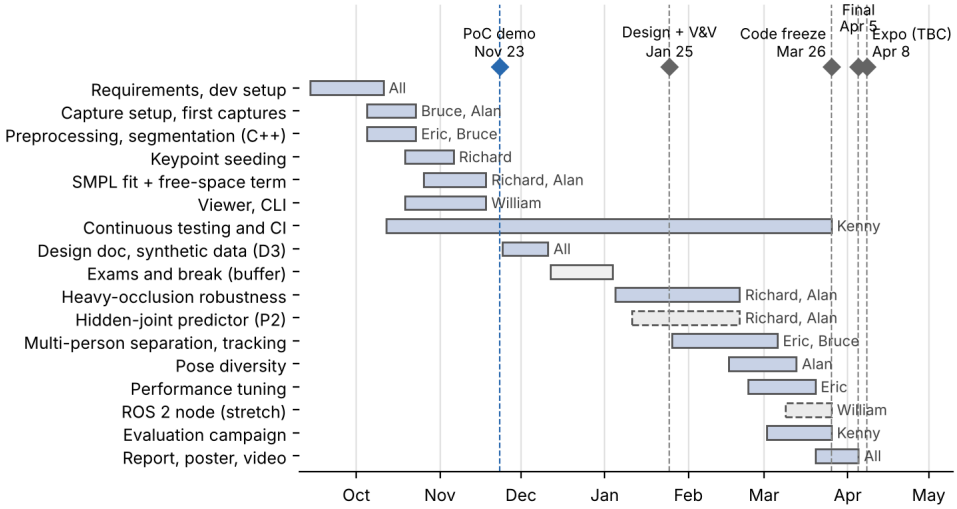


Figure 2. Work breakdown to the April 2027 Expo (dashed = stretch).

17.1 Open questions for Dr. Bone: must the output plug into ROS 2 or a specific planner, and with what latency? Is C++ required beyond the processing core? What is the exact Expo date?

Appendix A. Generative AI Usage

Claude (Anthropic) was used to analyze the technology stack and to help organize this document and check its consistency. The team made all technical decisions and is responsible for their accuracy.